CM3121 – Advanced Organic Chemistry

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Conjugate Addition

Readings: Clayden 2nd ed; Chapter 22 and 25 (Michael acceptors as electrophiles)

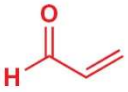
Outlines

I. Conjugate addition (1,4-addition) with
   Amine, thiol, cyanide as nucleophiles
   Organocopper reagents
II. 1,4 addition vs 1,2 addition
III. Conjugate additions with enolates
IV. Iminium ion catalysis
V. Conjugated addition in the presence of β-leaving group and Nucleophilic aromatic substitution S_NAr

Conjugated systems

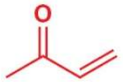
Conjugated systems

α,β-unsaturated aldehyde (an enal)



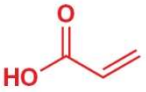
propenal
(trivial name = acrolein)

α,β-unsaturated ketone (an enone)



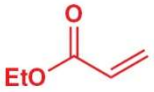
but-3-en-2-one
(trivial name = methyl vinyl ketone)

α,β-unsaturated acid

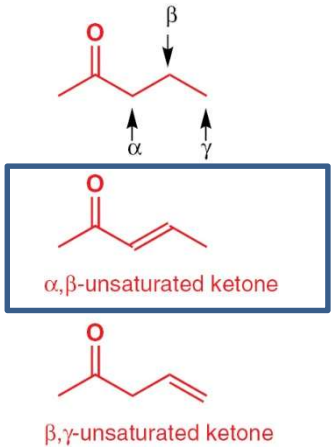


propenoic acid
(trivial name = acrylic acid)

α,β-unsaturated ester



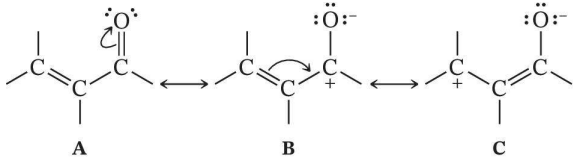
ethyl propenoate
(trivial name = ethyl acrylate)



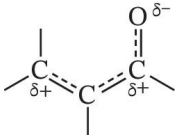
Reversed reactivity of the double bond (electron-deficient):



Resonance show positive charge (polarization)



Spectroscopy:

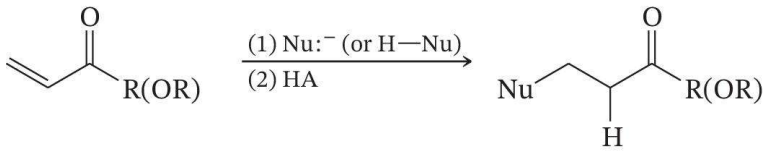


IR for simple ketone C=O 1715 cm^{-1}
conjugated C=O 1675 cm^{-1}
Compared with: C-O 1050 cm^{-1}

^{13}C NMR β-carbon 140 ppm (normal C=C is 120 ppm)

I. Conjugate addition (or Michael addition)

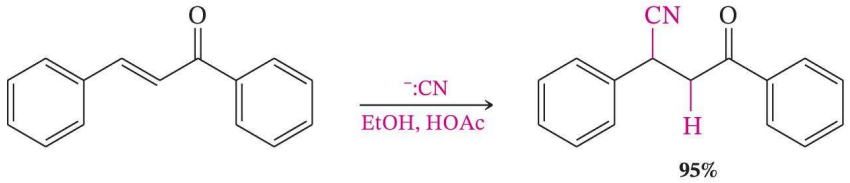
General reactions



Nucleophiles: CN^-, RNH_2, RSH, R_2CuLi, enolate, etc

Examples



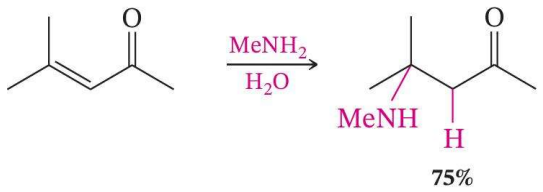


- Alkenes act as electrophile.
- Nu and H add across C=C double bonds.
- Must have a conjugated EWG (COH, COR, COOR etc) adjacent to the alkene (conjugated system).

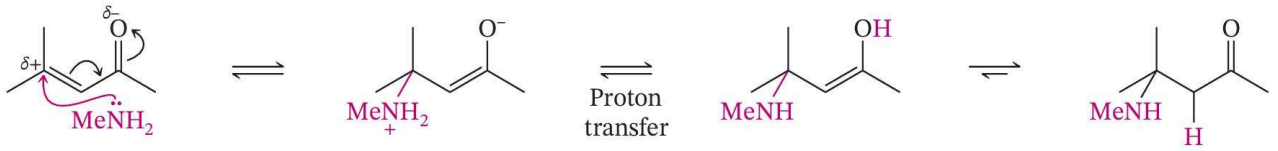
1.1 Addition of amines

Primary or secondary amines add to α,β-unsaturated carbonyl compounds yield β-amino carbonyl compounds

Examples



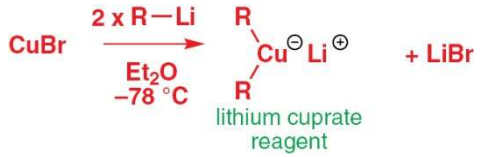
Mechanism:



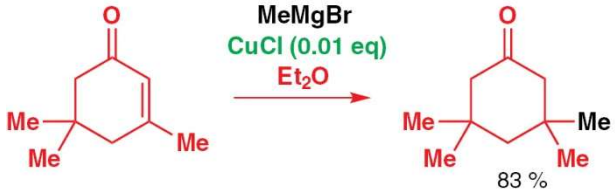
1.2 Addition of organocopper reagents

- Organocopper reagent: R_2CuLi (lithium cuprate) (The exact structure is more complex, tetramer)
  Cu (1.9) is much less electropositive than Mg (1.3). Thus C-Cu bond is less polarized than the C-Mg bond, giving the C atom less a partial negative charge.
- Organocoppers are softer than Grignard reagent.

Formation of organocopper reagents:



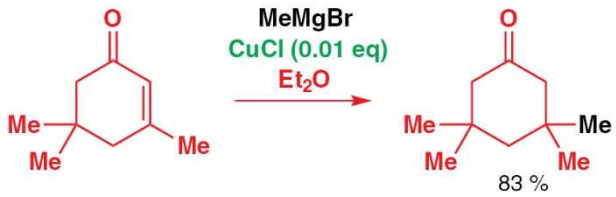
Not stable, should be used immediately



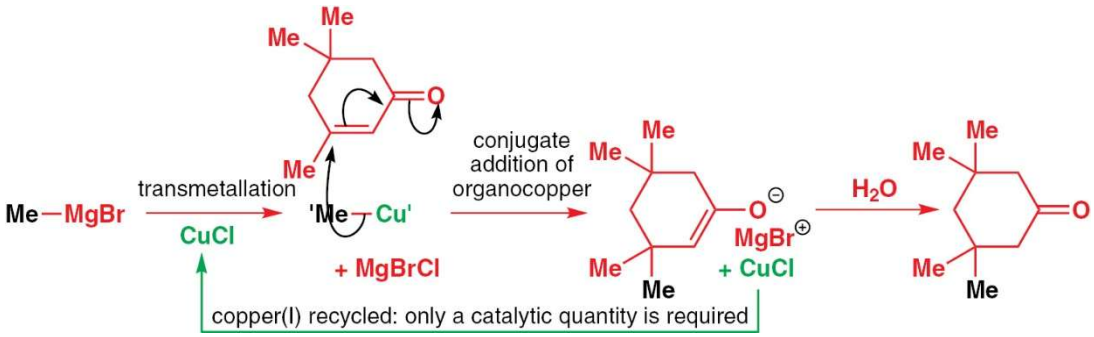
(In situ formation of organocopper using Cu catalyst)

Addition of organocopper reagents

- Conjugate addition



- Mechanism

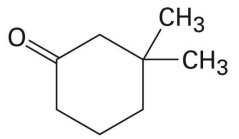


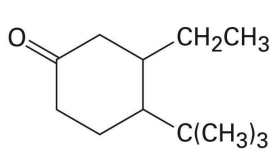
Without CuCl, addition of Me-MgBr to C=O takes place instead.

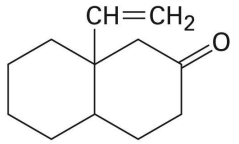
Practice

- How to use conjugate addition of organocuppors to prepare the following compounds?

(a) $CH_3CH_2CH_2CH_2CH_2C(=O)CH_3$

(b) 

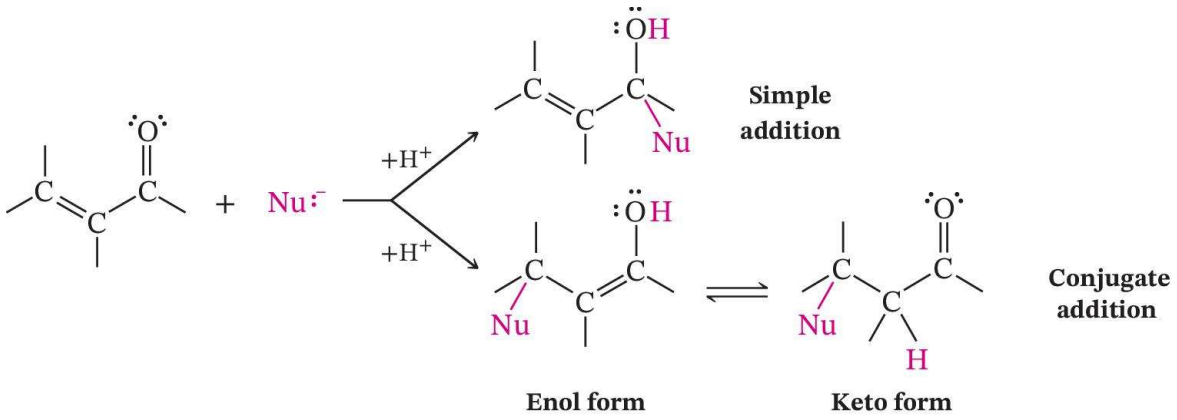
(c) 

(d) 

(consider only intermolecular conjugated addition reactions)

II. 1,2 vs 1,4

Addition takes place either directly at the carbonyl (simple addition, or 1,2-addition) or at the β carbon (conjugate or 1,4 addition).

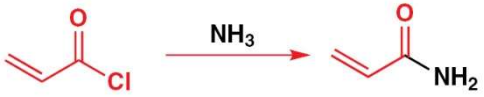


1,2 vs 1,4

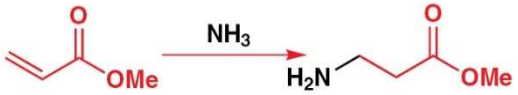
Which reaction (1,2 or 1,4) depends on:

1. The functional groups of the α,β-unsaturated systems

1,2



1,4

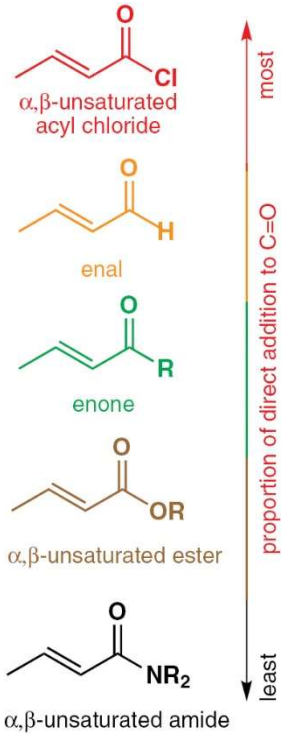


1,2



1,4 followed by 1,2

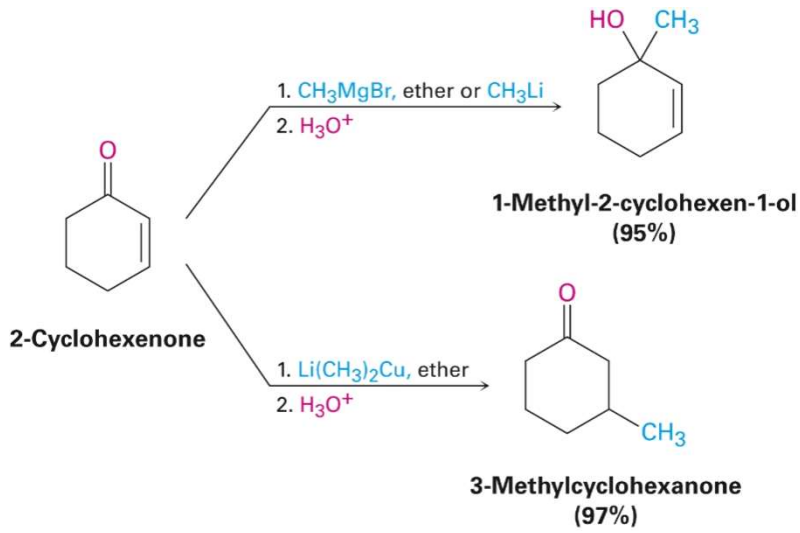




2. The Nucleophiles
3. The reaction conditions

Organocopper vs Grignard reagent

The nucleophiles



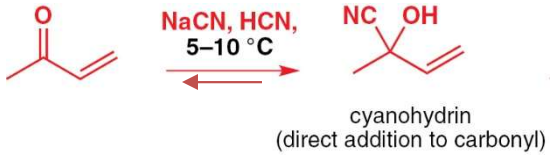
1,2 vs 1,4: nucleophiles

Hard Nucleophiles tend to react at the carbonyl carbon (hard electrophile) (1,2-addition)
Soft Nucleophiles tend to react at the beta carbon (soft electrophile) (1,4-addition)

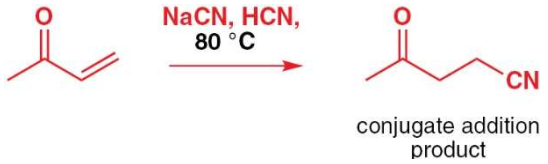
Hard Nucleophiles	Borderline	Soft Nucleophiles
F-, OH-, RO-, Cl-	N_3-, CH-,	I-, RS-, RSe-,
H_2O, ROH, ROR, RCOR'	RNH_2, RR'NH,	RSH, RSR', R_3P,
RMgBr, RLi,	Br-	Alkene, aromatic rings
		R_2CuLi

1,2 vs 1,4: reaction conditions (Cyanoaddition to methyl vinyl ketone)

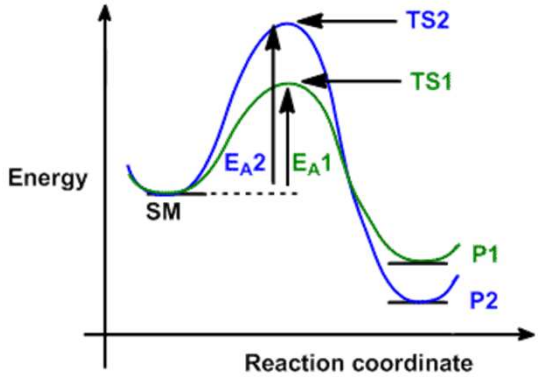
1,2 product form at low temperature (faster reaction, kinetic product)
1,4 product form at higher temperature (more stable product)

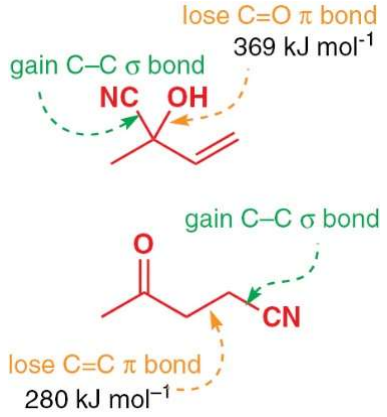


kinetic product (P1)



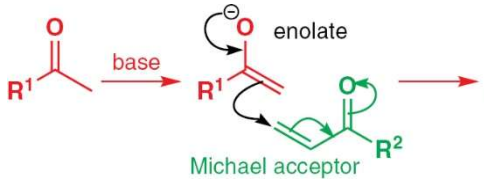
thermodynamic product (P2)



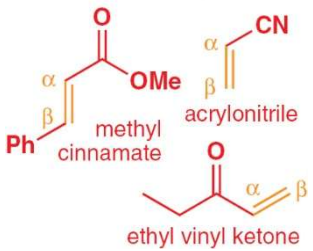


III. Addition of enolates

Conjugate addition

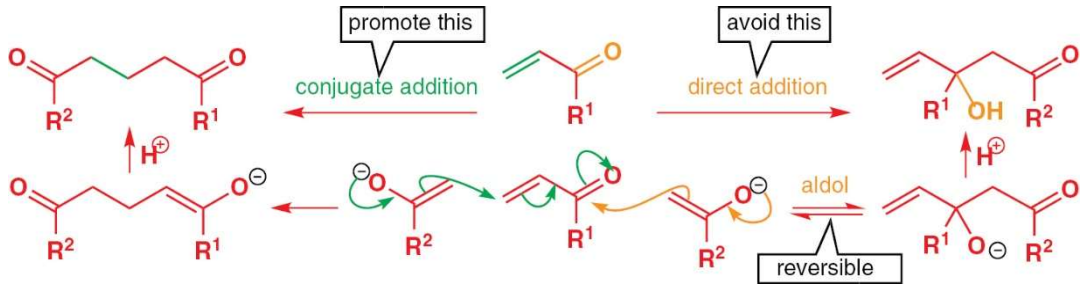


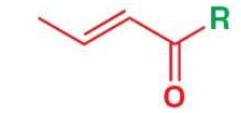
some Michael acceptors...



Addition of enolates

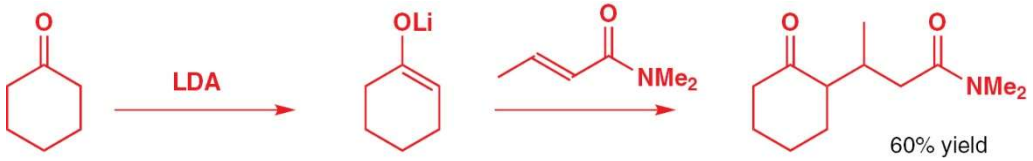
Competition between conjugated addition and aldol reaction

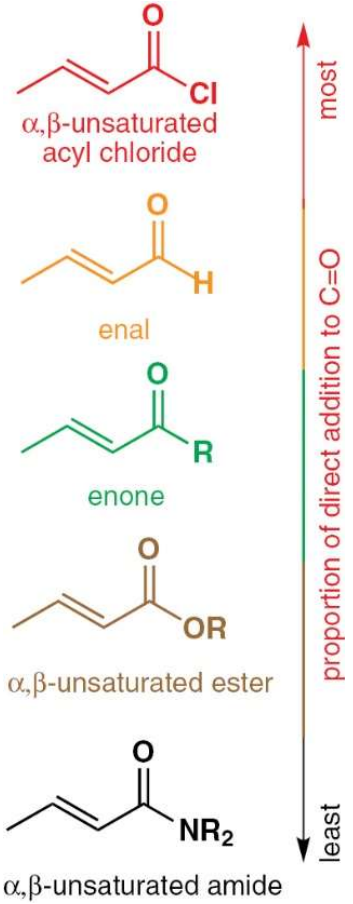




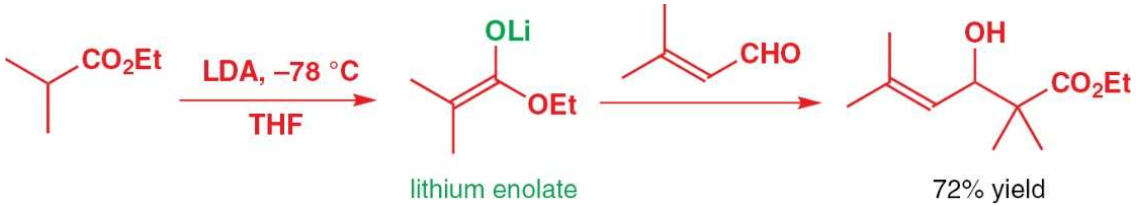
R = H < alkyl < OR < NR_2

increasing tendency
to conjugate addition



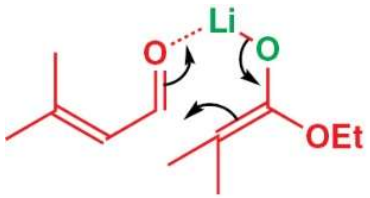


Addition of enolates (1,2 vs 1,4)



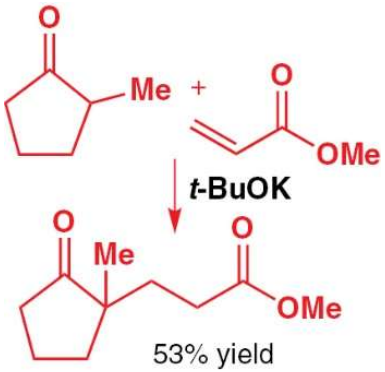
lithium enolate

72% yield

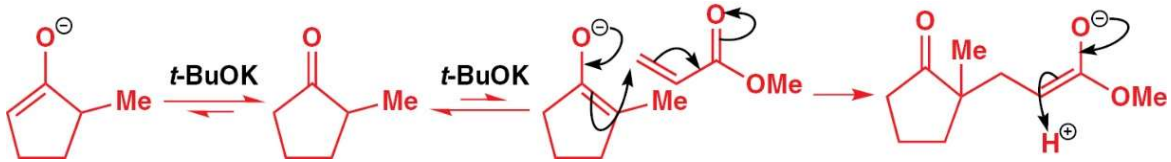


cyclic mechanism for ester aldol reaction

Lithium ion: coordinates to aldehyde and promotes direct aldol addition



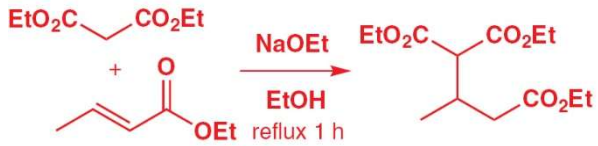
53% yield



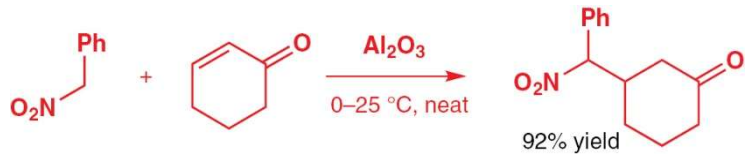
Thermodynamic enolate

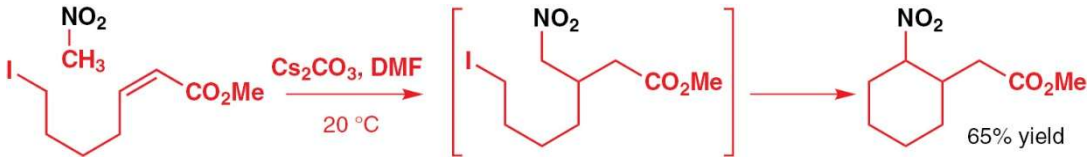
Stable enolate undergo conjugate additions

1,3 dicarbonyl compounds



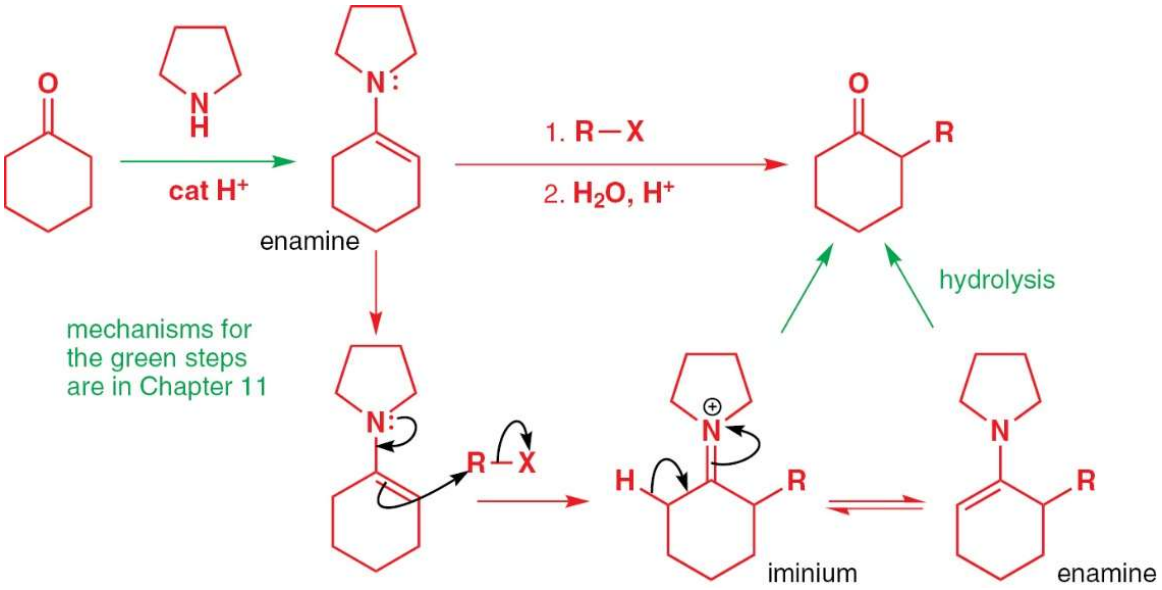
Nitroalkanes

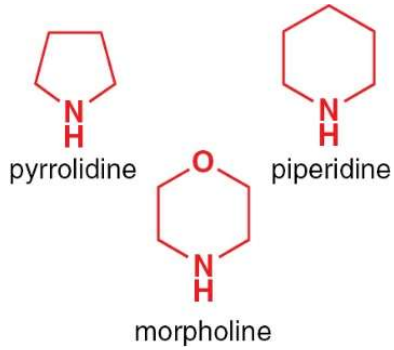




3. Enamines

Enamines are formed when aldehydes or ketones react with secondary amines.





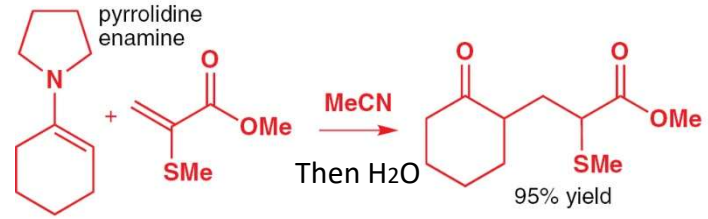
secondary amines

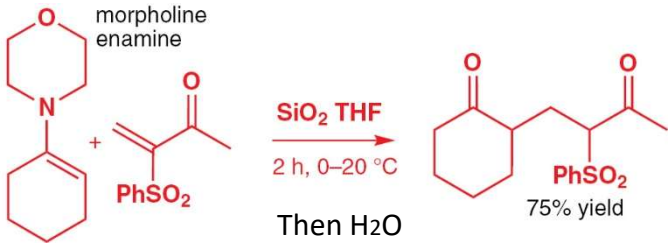
Imine and Enamine are tautomers

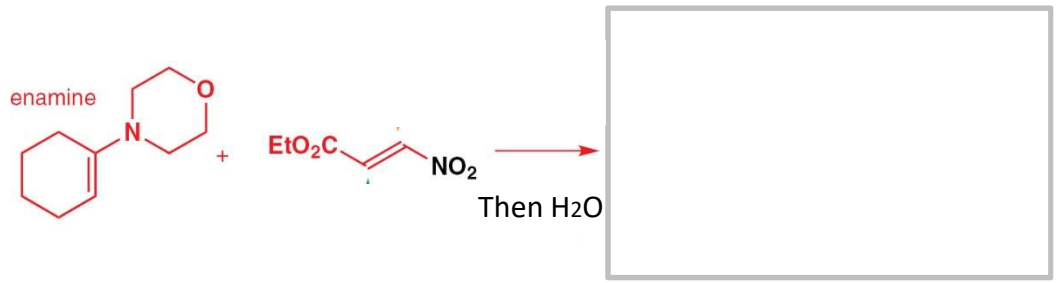
Recall: Reductive amination

Conjugate addition with enamines

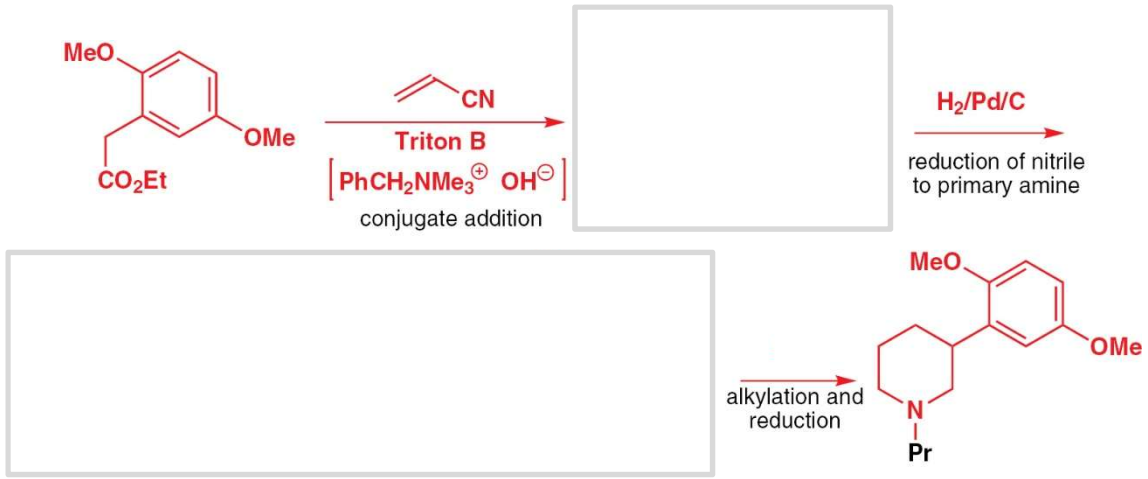
Enamine are neutral and soft nucleophiles: reliable for conjugate addition





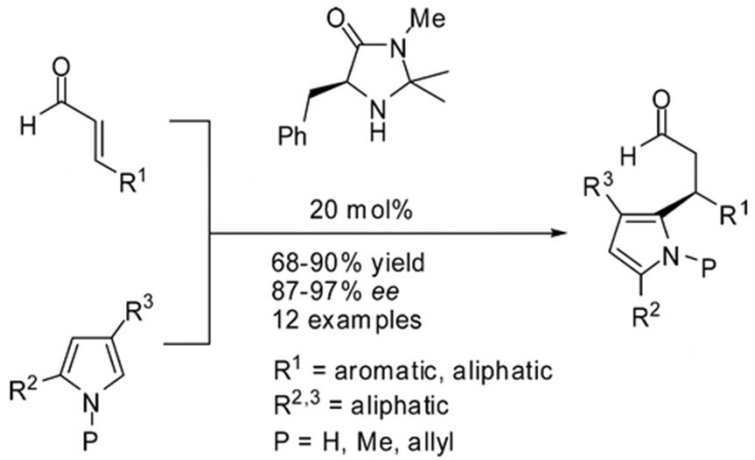


Synthesis application

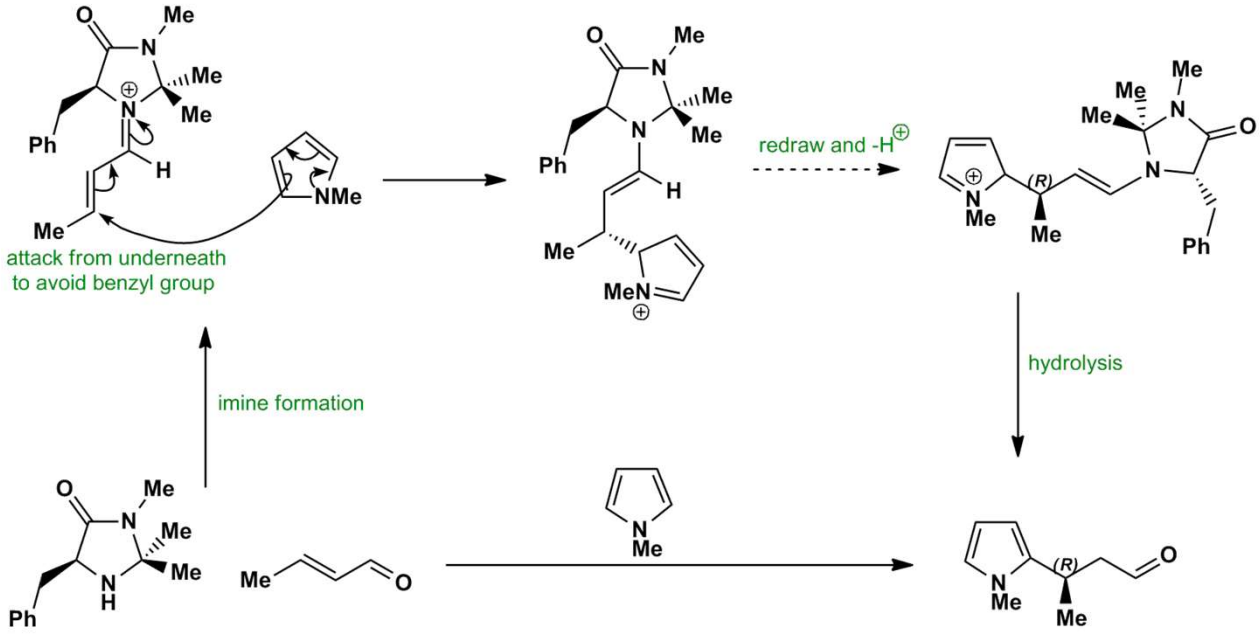


A dopamine antagonist
(dopamine antagonists found use in treating schizophrenia, bipolar disorder, and stimulant psychosis)

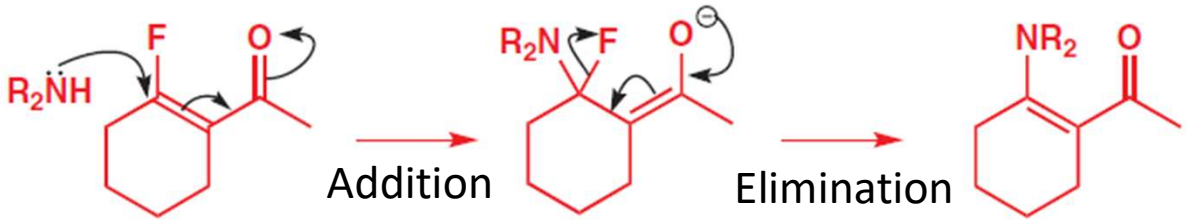
IV. Iminium ion catalysis



The iminium ion is more electrophilic than a neutral carbonyl compound

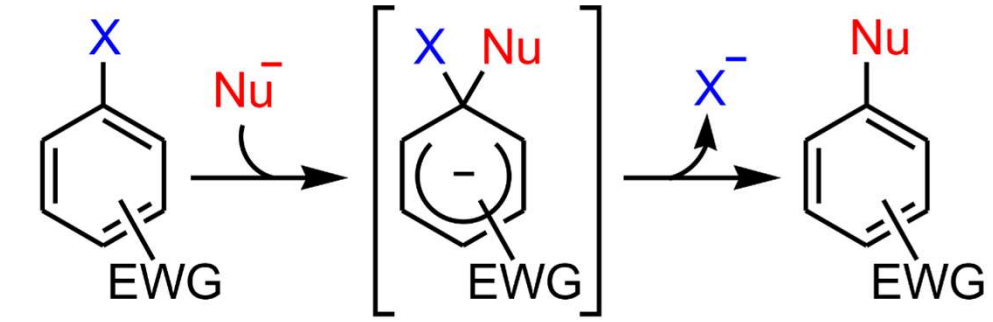


V. Conjugated addition in the presence of β-leaving group (e.g., F)

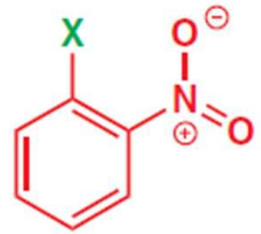


Addition-Elimination mechanism (substitution of β-leaving group)

Nucleophilic aromatic substitution S_NAr

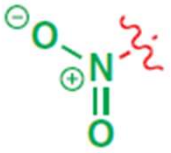


Nucleophilic aromatic substitution S_NAr

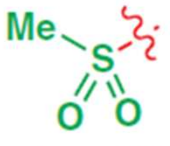


F >> Cl ~ Br >> I

F is the best leaving group



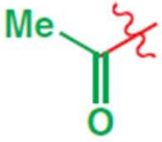
k_{rel} = 1



k_{rel} = 1/18



k_{rel} = 1/32

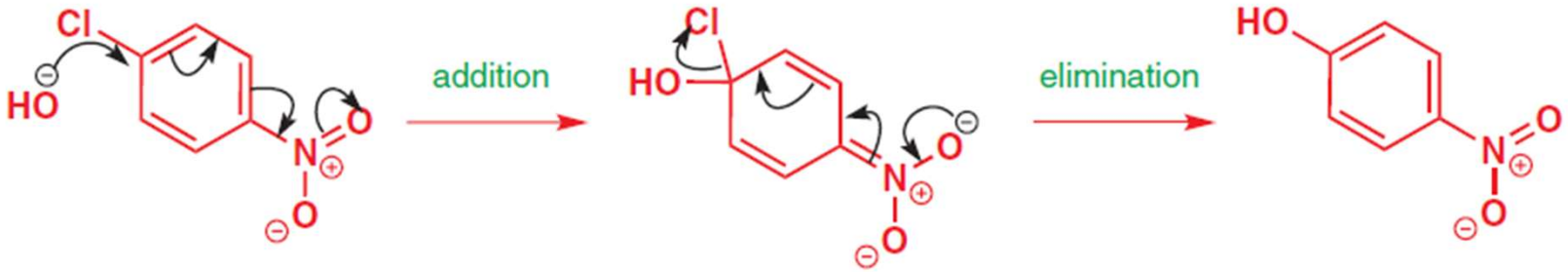


k_{rel} = 1/80

Strong EWG is needed

The leaving group should at the ***ortho-* or *para-* position** of the EWG





Summary

- Nucleophilic conjugate addition
  - 1,4 vs 1,2 addition
- Addition of organocopper reagents
  - RMgBr + CuCl (cat.)
- Addition of enolates and enamines
  - 1,5-dicarbonyl products
- Iminium ion catalysis
- Conjugated addition in the presence of β-leaving group
  - Nucleophilic aromatic substitution S_NAr

